



Investigating the effect of SiC particles on the physical and thermal properties of Al6061/SiC_p composite

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ABSTRACT

In the present study, the effect of volume fraction of SiC reinforcement particles on the physical and thermal properties of Al matrix composite (6061) was investigated. In this regard, composite powders with different reinforcement contents were prepared using hot press molding. It was seen that with increasing the amount of the reinforcement, the density of the samples decreased, while the porosity increased. Microhardness tests indicated that there was a direct relationship between the hardness and the amount of the reinforcement. SEM images confirmed the proper distribution of the reinforcing particles in the matrix. Dilatometry tests exhibited the decline of coefficient of thermal expansion by improving the amount of reinforcement. XFA test was also used to determine the thermal conductivity of the samples and it was seen that with increasing SiC content in the composite samples, the thermal conductivity declined to 61.5 W/m.K due to enhancement of crystalline defects in the phase boundaries. Finally, the experimental results were compared to thermoelastic models.

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1. Introduction

Al/SiC composites are of the most well-known composites for their high thermal conductivity, low thermal expansion coefficient along with good wear resistance that play an important role in modern technology [1–3]. By changing the reinforcement size, shape and content in these composites, the desirable physical and thermal properties can be obtained [4]. Although Al/SiC composites are currently the subject of many investigations, less attention has been attended to their physical and thermal properties, and in particular their dimensional stability. In electronic industries, it is vital to simultaneously reduce the size of circuit components and improve the power density. Thus, there is always an ever-increasing need for new high strength materials with high thermal conductivity and low thermal expansion coefficient to maximize the heat dissipation and minimize the thermal shock, respectively [1,3]. In the following different features of aluminum matrix composites for thermal applications are discussed:

1.1. Thermal conductivity of the composites

There are several mechanisms for explaining the thermal conductivity or, in other words, the transfer of thermal energy within a solid object. Various carriers, such as electrons, phonons (atomic vibrations) or photons, are considered for the transfer of thermal energy in a solid. In non-metallic materials, phonon transfer is considered as the main mechanism for heat transfer, while in metals heat is almost entirely transferred with the assistance of electrons. In alloys and semiconductors, both mechanisms are responsible for transmission of thermal energy. In general, thermal conductivity depends on the distribution of phonons [5,6]. In fact, thermal resistance is due to various types of processes, such as phonon-phonon scattering, scattering within boundaries, scattering by structural defects, etc. In order to maximize the thermal conductivity, phonon scattering processes must reach to their minimum values [7]. Increasing the thermal conductivity of the composites depends on a large number of variables that causes different properties of the material.

1.2. The size and shape of the reinforcement particles

The effects of reinforcement shape, particle accumulation and distribution of reinforcement and matrix was investigated on the

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thermal conductivity. The obtained results from these studies indicate that composites containing high percentages of the micro-scaled reinforcing agents have a higher thermal conductivity than the composites, which are reinforced by nano-scaled particles. This can be attributed to the greater scattering of phonons from the surface of nanoparticles with higher surface area [5–8]. In previous studies, it was observed that the composites reinforced by plate-like particles have higher thermal conductivity coefficient in comparison with composites containing spherical reinforcements. It can be attributed to the particles morphology. In case of plate-like reinforcements, due to higher aspect ratio, there is a lower average distance between particles, which could be a reason for higher heat transfer. In addition, high aspect ratio leads to formation of thermal conducting networks with low diffusion thresholds that reduce the disruptive role of interfacial thermal resistance on thermal conductivity [8,9].

1.3. Reinforcement surface modification

Investigation into the effect of reinforcement surface modification was carried out to provide better interactions between matrix and reinforcement, and evaluate its effect on the thermal conduction coefficient. Generally, surface modification of the reinforcement improves the reinforcement stability in the matrix, since surface modification prevents particle agglomeration it could increase the mixability of the matrix and reinforcement only if the proper modifier is selected [9,10].

1.4. Suitable reinforcement orientation

The proper orientation of the reinforcement particles has a significant effect on the thermal conductivity and final properties of the composite. This is particularly important for non-spherical reinforcements such as carbon nanotube, graphite/graphene and two-dimensional reinforcements such as boron nitride. The disadvantages of the thermal conduction behavior of the one-dimensional reinforcements are not discussed for two-dimensional reinforcements which is due to the high thermal conductivity of the two-dimensional materials and high surface-to-volume ratio of planar particles. The important point is the proper orientation of these plains, because of the anisotropy of planar reinforcements, i.e. when they are arranged perpendicular to the C-axis, their thermal conductivity are several times higher than those which are arranged along the C-axis. Therefore, preparation of composites with a specific orientation reinforcement, causes the composite thermal conductivity to change very significantly especially in the case of planar reinforcements that are arranged horizontally or vertically in the matrix [9,11].

1.5. The effect of matrix/reinforcement interface

One of the factors affecting thermal conductivity of the composites is the type of matrix/reinforcement interface. In this regard, formation of new compounds affects the heat transfer mechanism. In a research conducted by Ruch et al., it was seen that after preparation of Al-7SiC/Diamond and Al/Diamond composites by two methods of squeeze casting (SQ) and gas pressure infiltration (GPI), thermal conduction of the mentioned composites increased due to reaction of the matrix and the reinforcement at the interface and formation of Al_4C_3 compound. While silicon carbide does not tend to react with diamond, this phenomenon is related to the low entropy level of this reaction [12].

1.6. The effect of manufacturing process

The composite manufacturing process affects the thermal conductivity. As a result, oxidation possibility, introduction of external contaminations, final porosity and density, and the formation of unwanted compounds, which can differ final heat transfer rate of the composite, are all depend on the preparation method. The researchers observed that by producing Al/diamond composites via SQ and GPI methods, the GPI method, due to protection of melt against the oxidizing agents by inert gas and the gas pressure, provide much less porosities and the samples with higher density. Thus, the thermal conduction of the composite increased significantly [12,13].

1.7. Coefficient of thermal expansion (CTE)

Thermal expansion behavior depends on various factors such as microstructure, matrix deformation and internal stress conditions. The value of α changes with temperature, and in practice measuring temperature should be reported along with CTE value. In addition, the value of α in crystalline materials depends on crystal directions and this difference is particularly significant in single crystals. In polycrystalline ceramic materials, α can be considered constant in different directions, but if there is preferential arrangement of grains, CTE would vary depending on the direction [14]. In the composites, CTE decreases with increasing the ceramic silicon carbide particle. Basically, the conditions of CTE are explained by Visco-Plastic changes. In other words, non-elastic deformation of the matrix can affect the reversible thermal cycles. For Al-4Cu/SiC composite, it is reported that with increasing SiC particles, the CTE of the composite reduced linearly. This can be attributed to the presence of ceramic particles with very low thermal expansion coefficient in composites. Furthermore, the investigations indicated that CTE increased with increasing the temperature, but this increase is not linear [15].

The thermal fatigue resistance of aluminum matrix composites decreases with addition of ceramic particles. The service temperature of these composites is between ambient temperatures up to 300 °C, which may also increase significantly during service. In general, the behavior of the thermal expansion of the composites is a result of several parameters such as chemical composition and stability of the phases, volume fraction of the reinforcing particles, distribution of the reinforcing particles in the composite and the created defects during the manufacturing [16]. In case of SiC-reinforced 2080-aluminum alloy matrix, it was observed that by increasing the volume fraction of ceramic particles with low thermal expansion, the expansion coefficient of the composite reduced [17]. Normally, CTE of the composites is the result of several different parameters of the material, such as the chemical composition and stability of the phases, the volume fraction of the reinforcing particles, the distribution of the reinforcement particles and the created defects during the production. Usually, during production processes of casting or powder metallurgy, internal stresses are induced, which causes an incompatibility in the thermal expansions of composite constituents. Therefore, the thermal history of composites is important [16].

Residual stress also have influence on CTE of the composites. Chang et al. investigated the aluminum composite reinforced by 40 vol% Al_2O_3 particles. The samples quenched in water displayed a lower thermal expansion than those which were cooled in the furnace. This was related to the presence of residual thermal stresses in the quenched samples. In addition, nucleation of microporosities and/or growth of pre-existing porosities at the matrix/particles interfaces cause a difference in the strain during temperature change that slightly increases the CTE value [18].

Table 1
Chemical composition of Al(6061) matrix.

Element	Si	Fe	Cu	Mn	Mg	Zn	Cr	Ti	Al
6061Al	0.85	0.7	0.31	0.13	0.75	0.25	0.33	0.09	Re.

In this study, the effect of SiC particle volume fraction on the thermal and physical properties of Al (6061)/SiC_p composite was investigated. The related equations and models were also studied.

2. Experimental procedure

2.1. Materials

In order to prepare the composite samples, SiC (45 μ m, 99.8%, Merck) and 6061 aluminum alloy (45 μ m, 99.9%, Merck) powders were used. The chemical composition and morphology of the powder particles are brought in Table 1 and Fig. 1, respectively. For uniform mixing of the powders, mechanical milling process was employed by a high-energy ball mill. Powder mixtures with different reinforcement contents (5, 10, 15 and 20 vol%) were milled at 600 rpm for 2 h. For all experiments, stainless steel cup containing 5 stainless steel balls with average diameter of 20 mm and ball-to-powder weight ratio of 10:1 were utilized [27]. 1 wt% stearic acid was used as process controlling agent to prevent the powder particles from agglomeration. Argon gas (99.99% pure) was used as milling atmosphere. The powder mixture was hot-pressed in a cylindrical die with 140 mm length and 7 mm inner diameter at 450 °C for 30 min under 500 MPa pressure. Additionally, 2 graphite disks with 5 mm thickness were placed on the both ends of the sample.

2.2. Microhardness and density

Microhardness of the sintered samples was measured by Leitz microhardness under 500 g load for 15 s. In order to increase the accuracy, microhardness tests were repeated on four different spots and the average value was reported. To determine the density and porosity percent of the samples, Archimedes method was used according to ASTM B311-08 standard.

2.3. Characterization

In order to characterize the microstructure, particle distribution, size and morphology of the powder particles scanning electron microscopy (SEM) (LEO model 435VP, Netherlands) was used. In this way, samples were coated with gold to increase their conductivity.

2.4. Thermal examinations

2.4.1. Dilatometry test

Dilatometry test was done by (BEHER model DIL801, Germany) on samples with 6 mm diameter and 14 mm length. Coefficient of thermal expansion and dimensional changes of the samples were measured in the temperature range of 25–450 °C with heating rate of 5 °C/min. This process was done for 3 heating and cooling cycles. The length tolerance of the diameter was considered 0.1 μ m.

2.4.2. Xenon flash thermal diffusivity analysis

Thermal diffusivity and thermal conductivity coefficient (K) were obtained using xenon flash thermal diffusivity analyzer (XFA model 500, Germany) based on ASTM E1461 standard. Samples with 12.7 mm diameter and 5 mm thickness were exposed to high-intensity energy pulse. The energy of the pulse is absorbed by the samples, thus thermal diffusivity of the sample is calculated based on the thickness and the time required for temperature elevation of the sample surface. By having the thermal diffusivity, heat capacity (C_p) and density of the samples at the test temperature, thermal conductivity of the samples (from 25 to 500 °C) were determined.

3. Results and discussion

3.1. Distribution of reinforcement particles in Al matrix

Fig. 2 shows the SEM micrographs of samples containing 5, 10, 15 and 20 vol% SiC. As is shown in these micrographs, a uniform distribution of the reinforcement in the Al matrix was obtained from the mixing operation. Fig. 3 shows a magnification of the matrix/SiC particles interface. As shown in the figure, aluminum powder particles are well bonded to each other, no crack is observed at the matrix/reinforcement interface, and it was found that press force, temperature, and sintering time were appropriate during the manufacturing process.

3.2. Hardness measurement

Fig. 4 shows that the hardness of the composite samples improved with increasing SiC volume fraction. This can be attributed to the presence of hard silicon carbide particles in the soft aluminum matrix, which increases the density of crystalline defects, especially dislocations. Furthermore, silicon carbide particles act as obstacles to the motion of dislocations [19]. According to the graph, the hardening rate from declined with increasing the amount of reinforcement when the reinforcement content increased from 15 vol% to 20 vol%. It can be said that by increasing the amount of silicon carbide, compression of the powder becomes more difficult, resulting in porosity between the particles, which

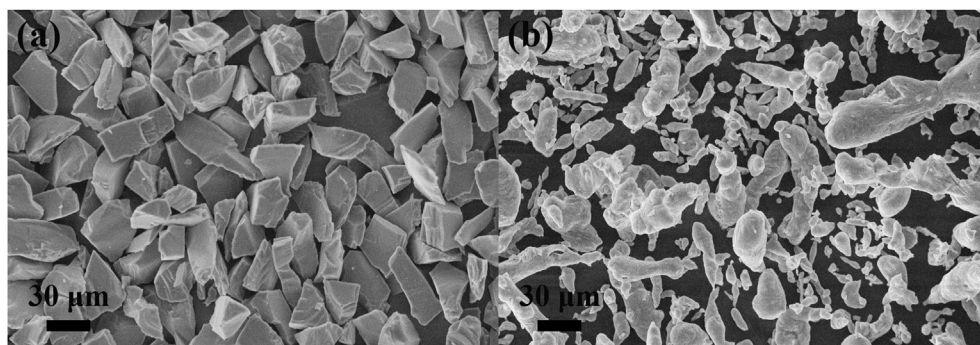


Fig. 1. SEM micrographs of: (a) SiC powder and (b) Al (6061) powder.

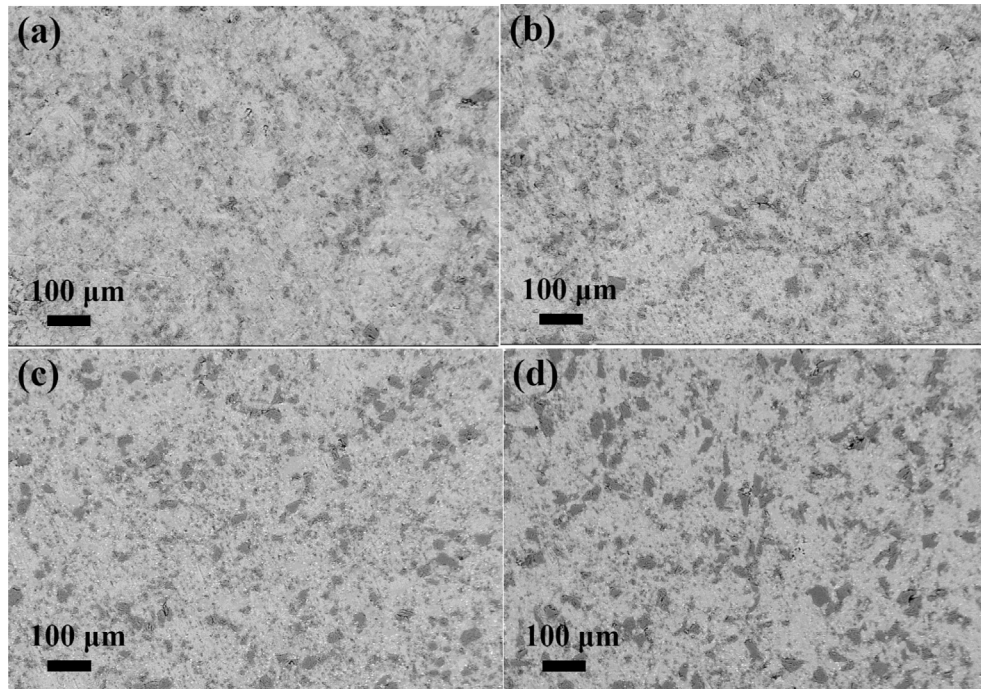


Fig. 2. SEM micrographs of the samples after pressing and sintering process: (a) 5%SiC, (b) 10%SiC, (c) 15%SiC and (d) 20%SiC.

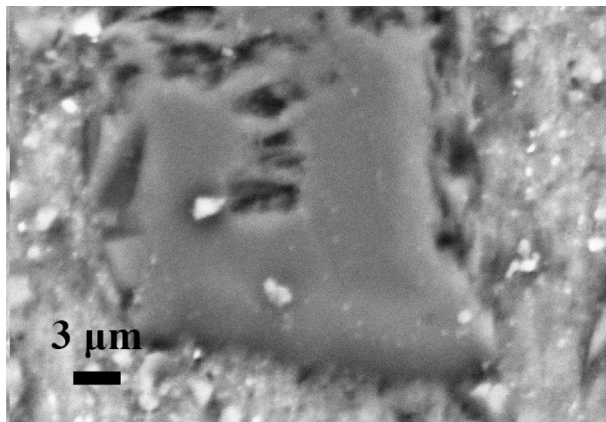


Fig. 3. SEM micrograph of matrix/reinforcement interface.

itself reduces the hardening rate. Also, the phenomenon of agglomeration can be noted, which leads to a reduction in the hardness rate.

3.3. Density of the prepared specimens after sintering

Fig. 5 shows the relative density changes and the porosity of the samples in terms of SiC content. As can be seen, the samples density reduced with increasing SiC content. Since SiC is a hard material, the compressibility declined by rising the SiC percentage. Due to the large difference between the melting points of the matrix and the reinforcement, with enhancing the SiC content, the sintering process of the powder was disrupted, resulting in formation of porosities and cracks in the composite. Other reason contributing to the reduction of density is agglomeration, which increases with improving the SiC content. As a result, the formed small porosities inside agglomerates and large porosities between them increased the porosity and reduced the density [20]. Removing these

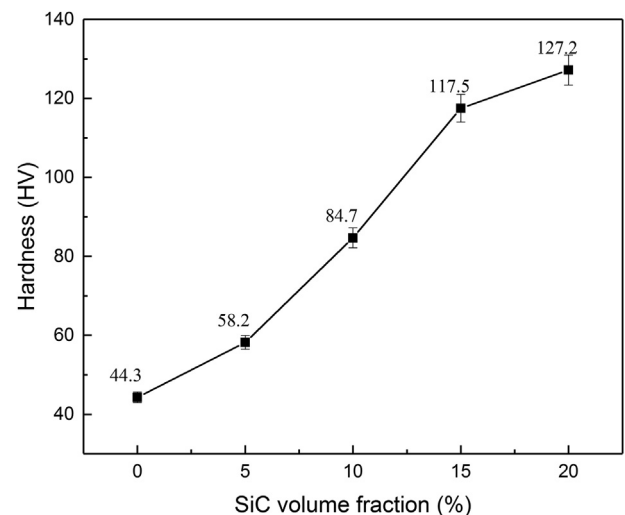


Fig. 4. Microhardness variation of Al (6061)/SiC_p composite samples with respect to SiC content.

porosities requires applying high pressures and elevating the heating time. This enhancement of temperature causes the grain growth and loss of mechanical properties [21]. Low SiC density compared to aluminum is also another factor in reducing the density of the composite sample, i.e. by increasing the amount of reinforcement, density of the samples reduced. By comparing the apparent and the theoretical densities in Fig. 6, it is apparent that at first the difference of the two values was low, but by enhancing the amount of reinforcement the difference increased. This is because the effect of porosity is not considered in calculating the theoretical density. Due to the lack of powder compaction, samples containing large amounts of reinforcement had higher porosities in the way that greater difference was appeared between experimental and rule of mixtures values.

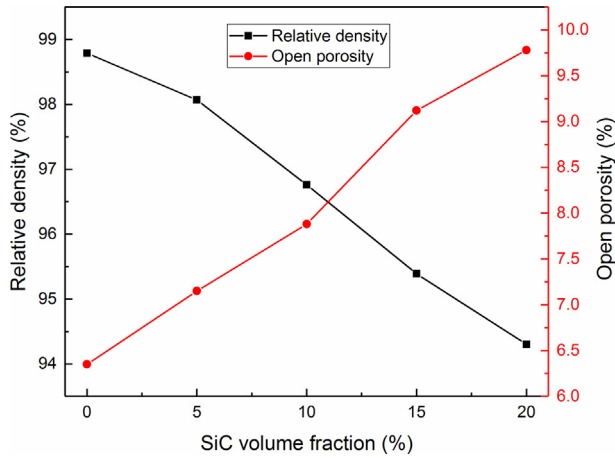


Fig. 5. Change of relative density and open porosity for sintered composite samples.

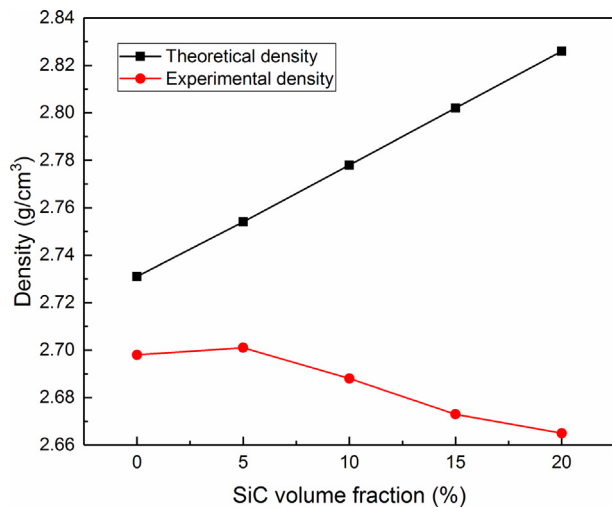


Fig. 6. Comparison of theoretical density and apparent density of Al (6061)/SiC_p after sintering.

3.4. Coefficient of thermal expansion

Thermal behavior of the samples contained different SiC contents was examined in a range of room temperatures up to 450 °C in three heating/cooling cycles. Fig. 7 shows the CTE results in terms of the volume fraction of SiC particles. As can be seen, the results are in good agreement with the line with the equation $y = 24.3 - 0.54V_p$. The result showed the CTE relation with the particle volume fraction, regardless of particle size, which was obtained by thermal analysis. Researchers found that the size of the reinforcement particles do not have a dramatic effect on CTE, but the amount of reinforcing particles is a very effective factor in the thermal behavior of the composites, as it was observed in a study on Al/TiC composites. The increase in the particle size of the reinforcement from 0.7 mm to 4 mm has a slight change in the CTE (less than 10%) composite, which is due to the distortion of the network in the surface areas [22]. As shown in Fig. 7, by increasing the amount of reinforcement, CTE reduced, which can be attributed to two factors: the first is the difference in the thermal properties of aluminum and SiC. In fact, due to lower CTE of ceramics in comparison with aluminum, a decrease in CTE value occurs if added to the matrix. The second can be attributed to the irregular shape of the SiC particles with respect to fully-spherical morphology, which

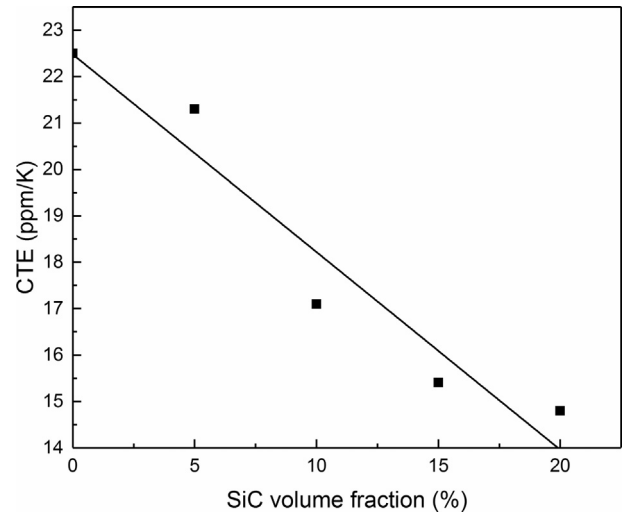


Fig. 7. Trend of CTE vs. reinforcement content in temperature range of 100–450 °C which was matched to line equation of $y = 24.30 - 0.54V_p$.

induces different thermal stress fields, reduces the linear expansion of the matrix and consequently decreases the linear expansion of the composite.

3.4.1. Comparison of dilatometry analysis with the proposed models

To study the thermal behavior of materials and calculate the CTE, two models were presented that the simplest one of them is the rule of mixtures Eq. (1).

$$\alpha_c = V_p \alpha_p + V_m \alpha_m \quad (1)$$

Where, m and p represent the matrix and reinforcement, respectively. Advanced models are based on thermoelastic theories. The Schapery model calculates the upper and lower bands of the CTE. A specific expression of it is given as Eq. (2) [15].

$$\alpha^{(+)} = \alpha_p + (\alpha_m - \alpha_p) \left[\frac{k_m (K_p - K_c^{(-)})}{K_c^{(-)} (K_p - k_m)} \right] \quad (2)$$

Where, $K_c^{(-)}$ is composite bulk modulus that is calculated by the Hashin and Shtrikman model (Eq. (3)).

$$K_c^{(-)} = K_m + \left(\frac{V_p}{\frac{1}{K_p - K_m} + \frac{V_m}{K_m + \frac{4}{3}\mu_m}} \right) \quad (3)$$

Where, μ_m is the shear modulus of the matrix. By changing the m and p in the equation, Schapery upper band modulus can be calculated. By replacing the bulk modulus in Eq. (2), CTE of the composite can be obtained. The upper band of the Schapery model is expressed simultaneously with the Kerner model. Fig. 8 shows the experimental and theoretical results of the proposed models for the CTE of Al/SiC composite in terms of reinforcement volume fraction. By comparing the fitted lines in the figure above, empirical data decreased more quickly than the Schapery models. In fact, due to the high difference of SiC and aluminum CTEs, the dislocation density increases at the matrix/reinforcement interface that leads to stress formation. As a result of this phenomenon, the effect of induced thermal strains is not considered in the proposed

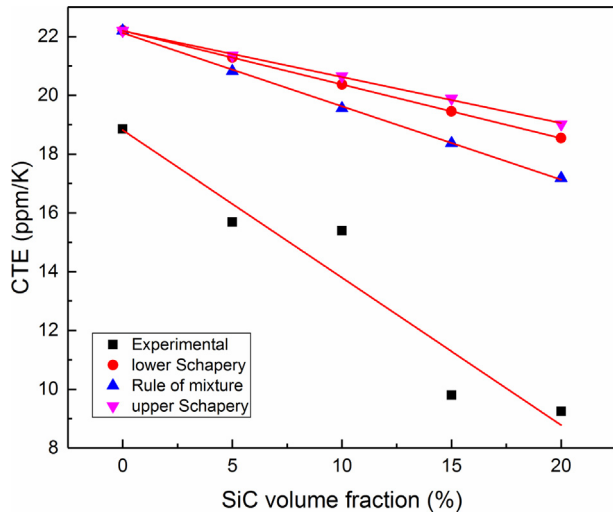


Fig. 8. Comparison of CTE experimental and theoretical results (line of $y = 19.88 - 0.8V_p$) vs. reinforcement volume fraction in 100–150 °C temperature range.

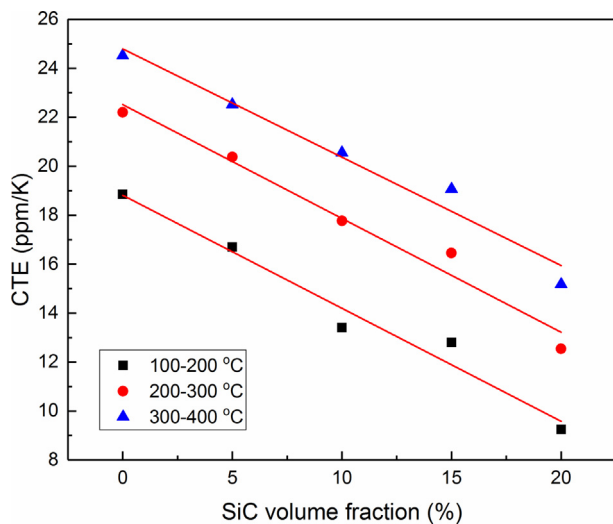


Fig. 9. CTE experimental results vs. reinforcement volume fraction in different temperature ranges for Al/SiC_p composite samples.

thermoelastic models. Therefore, in the graph of experimental data, CTE was changed more quickly.

Fig. 9 shows that by elevating the temperature, the slope of the lines, which were fitted to theoretical and experimental results increased. This indicates the increase of CTE by increasing the temperature. It should be noted that the improvement of CTE with temperature is due to approximate increase of SiC CTE at high temperatures.

Fig. 10 shows the CTE variations of the sample containing 5 vol% SiC in three heating cycles. As it can be seen, CTE variations was not significant at successive heating and cooling cycles, which indicates the stable thermal behavior of the samples. CTE changes at temperatures above 400 °C is because of sudden change of SiC CTE.

Fig. 11 displays the thermal behavior of Al 6061 alloy and the composite specimens with different reinforcement contents in the first heating cycle. By more precise observation of the figure it can be seen that CTE decreased by increasing the reinforcement. The remarkable point in this diagram is the change of samples thermal behavior at temperatures higher than 400 °C. The cause of this

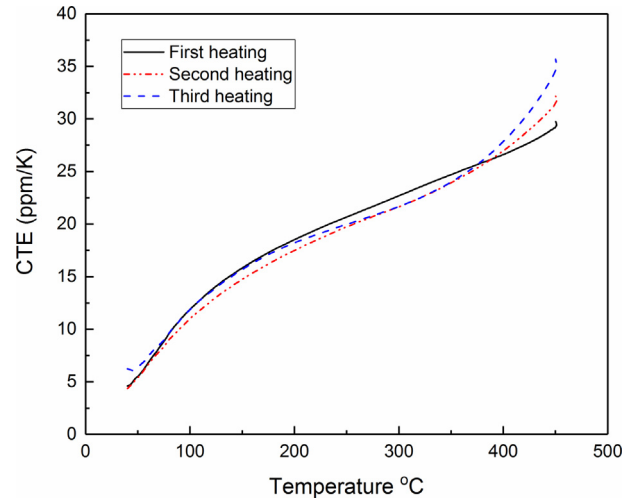


Fig. 10. Dilatometry results for relative CTE of Al-15%SiC composite sample in 3 sequential heating cycles with heating rate of 5 °C/min.

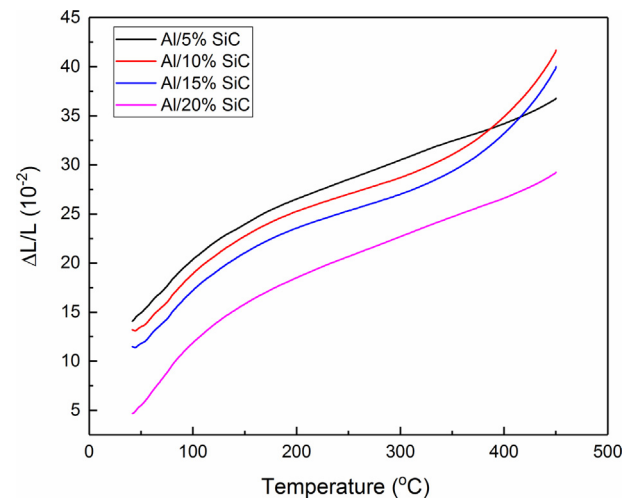


Fig. 11. Comparison of thermal behavior of Al (6061) and Al/SiC_p composite samples containing 5%, 10%, 15% and 20% of reinforcement vs. temperature.

phenomenon can be related to release of elastic-plastic stresses resides at the matrix/reinforcement interface, which causes a permanent expansion in the sample at high temperatures. These tensions are released during cooling. In fact, releasing of these stresses and changing of SiC CTE at this temperature results in this behavior.

Another noteworthy point is the high difference of the sample containing 20 vol% reinforcement with respect to other composite samples. In samples with high SiC contents, SiC significantly affects the thermal behavior; the slope of this variation is higher at low temperatures due to the difference of matrix and reinforcement CTEs. At temperatures above 400 °C, because of surface melting of aluminum particles, CTE variations have undergone severe changes.

The experimental CTE curve of Al-15%SiC composite sample in the heating/cooling cycle is shown in Fig. 12. In this curve, the experimental results were compared to Turner model and it is evident that at temperatures below 150 °C, the experimental curve is placed below the Turner model. The presence of SiC particles in the composite has led to reduction of CTE. By comparing the experimental CTE curve and the SiC CTE, it can be observed that at low temperatures, CTE of the composite samples is not a function of

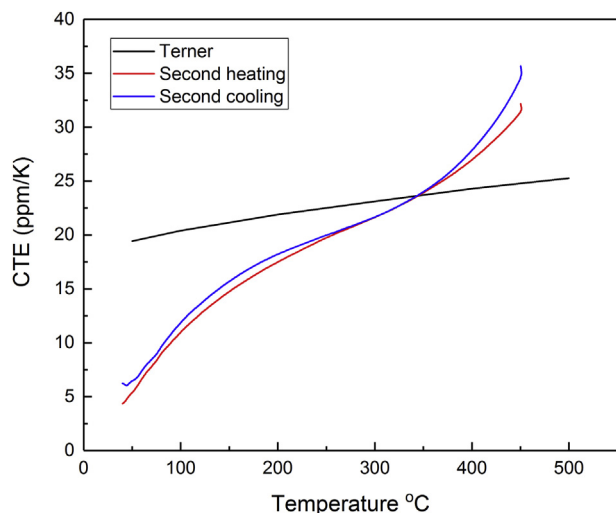


Fig. 12. Comparison of Turner Model and experimental results for CTE of Al-15%SiC_p in the second heating/cooling cycle.

matrix CTE. In other words, CTE of the matrix cannot help determine the composite CTE, which is, in fact, a failure for thermoelastic models.

3.5. Thermal conductivity

The rate of heat transfer was investigated by the XFA test. This test was carried out at ambient temperature with heating rate of 20 °C/min. The thermal capacity of the samples was also calculated by Eq. (4).

$$C_{theor} = \frac{C_{Al}V_{Al}\rho_{Al} + C_{SiC}V_{SiC}\rho_{SiC}}{\rho_{theor}} \quad (4)$$

Where, C is specific heat, V is the volume fraction and ρ is density.

After calculation of the samples heat capacity, heat transfer coefficient (k) can be calculated by means of Eq. (5).

$$k = D \times \rho_{theor} \times C_{theor} \quad (5)$$

Where, D is the thermal diffusivity of the composite.

Table 2 shows the experimental values of CTE and the theoretical heat capacity of the composite samples for samples with different reinforcement contents. If a simple composite model is assumed for the uniform distribution of spherical SiC particles in the aluminum matrix, the effective thermal conduction is obtained according to Maxwell equation. According to Maxwell theory, the overall thermal conductivity of the composite is due to the thermal conductivities of the reinforcing and the matrix particles and amount of each of these two, and this value is independent of particle size [23].

In Fig. 13, the graph of CTE variations vs. reinforcement content

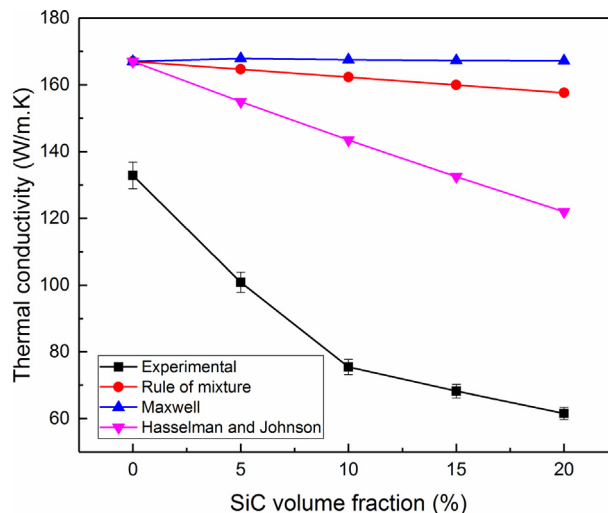


Fig. 13. Comparison of thermal conduction coefficient of Al (6061)/SiC_p composite samples (experimental results, Maxwell Theory and rule of mixtures).

is depicted. In this diagram, it can be seen that the thermal conductivity of the composite is dependent on the reinforcement content and by increasing the amount of reinforcement, heat transfer rate decreased. This means that the thermal conductivity of the composite samples depends heavily on the amount of reinforcement, the interface and the amount of remaining porosities in the samples. It can also be acknowledged that the interface between the matrix and the reinforcement acts as a barrier to heat conduction and by increasing the reinforcement content, the amount of interfaces and consequently the barriers to heat transfer improved, so the rate of heat conduction decreased. A notable feature of this diagram is the reduction of its slope by increasing the reinforcement content. In order to analyze this process, it can be stated that by increasing the amount of hard SiC phase at a constant time, SiC particles were less fractured during the mechanical alloying process. Therefore, higher milling time is required in comparison with the sample with lower SiC contents. For higher SiC contents, the particles become coarser and hence lower interface is formed, which leads to better thermal conductivity. In comparison with other studies, which used casting method for preparation of Al-SiC composites the quality of the interface of matrix/reinforcement in the present study was so much better since by hot press molding the temperature of the process was lower than casting method and the possibility of formation of unwanted compounds such as Al₄C₃ [24]. The presence of unwanted compounds can highly reduce the thermal conduction of Al matrix composites and lower the mechanical properties of the interface [25].

In Fig. 13, the experimental results of this study were compared to the results of Maxwell theory and rule of mixtures. As can be seen, in Maxwell model the only variable is the amount of heat flux that according to the results of Table 2, the difference in thermal diffusivity of samples with different reinforcement contents is low.

Table 2
Comparison of experimental results of thermal conduction coefficient and heat capacity for composite samples.

Sample	SiC volume fraction (%)	Heat capacity (J/kg.K)	Thermal diffusivity (cm ² /s)	Thermal conductivity (W/m.K)
1	0	892	48.11×10^{-2}	132.87
2	5	874.49	34.21×10^{-2}	100.87
3	10	866.92	23.84×10^{-2}	75.46
4	15	859.48	21.08×10^{-2}	68.24
5	20	852.16	18.35×10^{-2}	61.55

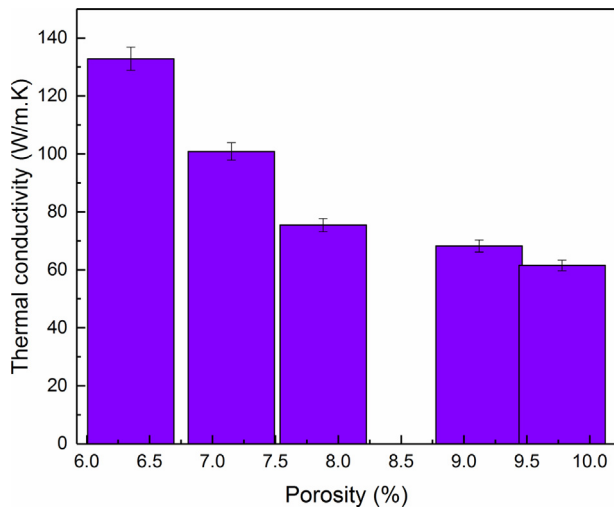


Fig. 14. The trend of thermal conduction coefficient variation vs. porosity for Al (6061)/SiC_p composite samples.

By inserting the values in Eq. (5), it is realized that the thermal conduction coefficient do not differ considerably and can be neglected. In fact, in Maxwell model, the effect of the thermal resistance of the phasic interfaces is not considered, which is similar to the results of rule of mixtures. It should be noted that the great difference of experimental results with the Maxwell model and the rule of mixtures is due to presence of the porosities and phasic interfaces, which these factors are not considered in theoretical models. In a recent study, it was found that the experimental results of thermal behavior of Al-graphite composite are consistent with Maxwell model. This difference between the investigations could be also attributed to the reinforcement type and isotropy. Moreover, no porosity was reported to be found at matrix/reinforcement interface. It is another important factor, which is highly effective in consistency with theoretical models, too [26].

The values of reinforcement particle size ($D = 45 \mu\text{m}$), reinforcement volume fraction, $h = 6.7 \times 10^7 \text{ W}(\text{m}^2 \cdot \text{K})$, $K_p = 120 \text{ (W/m.K)}$ and $K_m = 167 \text{ (W/m.K)}$ were inserted in Hasselman-Johnson and the graph of Hasselman-Johnson theory was plotted as shown in Fig. 13. As can be seen, the empirical results are in good agreement with theoretical data, indicates that in the Hasselman-Johnson model, the parameters of size, volume fraction, thermal resistance of the interfaces, etc. are considered and the results estimated by this model are close to the experimental results.

Fig. 14 shows the variations of CTE in terms of the porosity of the samples. Values of the horizontal axis were attained by the diagram of Fig. 5. It is deduced from the chart that the porosity of the samples increased with increasing the reinforcement content. It affects the heat transfer rate of the composite, so the existing porosity reduces the thermal conductivity of the composite, and in fact acts as an interface by creating thermal resistance. One of the possible reasons for this phenomenon is the tendency of porosities to accumulate at phase boundaries. By increasing the density of porosities at the interface areas, thermal conductivity decreases considerably. This phenomenon shows the reason for the difference between theoretical and empirical results.

Fig. 15 shows the variation of thermal conduction coefficient of Al (6061)/SiC composite with respect to CTE. This diagram shows the direct relation of these parameters with each other in a way that by increasing the thermal expansion coefficient, heat conduction of the samples increased. The lower the amount of reinforcement in the composite, the lower the density of dislocations at

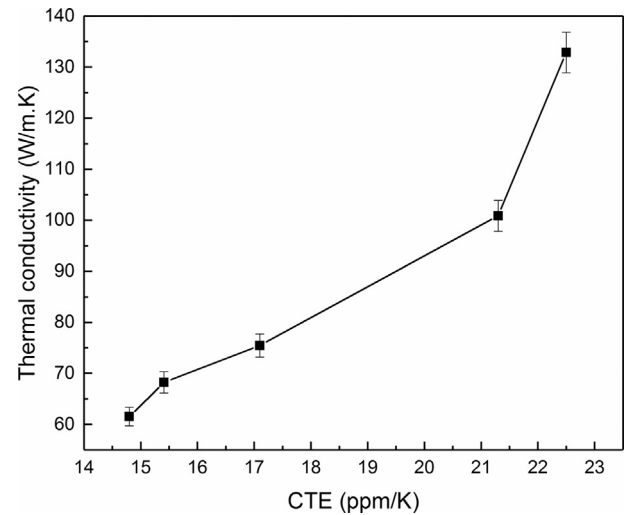


Fig. 15. Change of thermal conduction coefficient vs. CTE for Al (6061)/SiC_p composite samples.

the interfaces, which results in a higher atomic order in these areas. Therefore, it can be concluded that with increasing the SiC content in Al/SiC composite samples, the density of crystalline defects such as dislocations in the boundary regions increased and reduced the atomic order. Thus, according to Rayleigh's model for thermal conductivity analysis of composites; which considers the high heat conduction as a result of high atomic order, by increasing the amount of reinforcement the density of dislocations in the boundary regions increased. Subsequently, the thermal expansion and thermal conductivity values reduced.

4. Conclusions

- (1) The sample preparation conditions were selected as 30 min sintering time and 500 MPa pressure. Hardness improved by increasing SiC content. In a way that by adding 20 vol% SiC, the microhardness was reached to 127 HV. Due to the disruption of the powder compaction process and agglomeration, by enhancing the amount of SiC in the composite samples porosity increased and density declined.
- (2) The thermal expansion coefficient of the samples decreased linearly by elevating the SiC content, so that the CTE of sample containing 20 vol% reinforcement declined to 14.8 ppm/K.
- (3) The suggested thermostatic models were unable to analyze the CTE diagrams accurately, since they do not consider the created internal strains.
- (4) The prepared samples by the mentioned method had proper thermal stability in heating and cooling cycles.
- (5) At temperatures close to 400 °C, due to the release of thermoelastic stress and the sudden change of SiC thermal behavior, the CTE values of the samples improved significantly.
- (6) With increasing SiC content in the composite samples, the thermal conductivity decreased.
- (7) By increasing the SiC content, crystalline defects enhanced in phase boundaries. For this reason the thermal conductivity of the sample containing 20 vol% declined to 61.5 W/m.K.

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